Hence, $\theta_3 = \theta_1$. The slab displaces the beam's position, but the beam's direction remains unchanged.

Exercise 8-4: In the visible part of the electromagnetic spectrum, the index of refraction of water is 1.33. What is the critical angle for light waves generated by an upward-looking underwater light source?

Answer: $\theta_c = 48.8°$. (See EM.)

Exercise 8-5: If the light source of Exercise 8-4 is situated at a depth of 1 m below the water surface and if its beam is isotropic (radiates in all directions), how large a circle would it illuminate when observed from above?

Answer: Circle's diameter = 2.28 m. (See EM.)

8-3 Fiber Optics

By successive total internal reflections, as illustrated in Fig. 8-12(a), light can be guided through thin dielectric rods made of glass or transparent plastic known as ***optical fibers***. Because the light is confined to traveling within the rod, the only loss in power is due to reflections at the sending and receiving ends of the fiber and absorption by the fiber material (because it is not a perfect dielectric). Optical fibers are useful for the transmission of wide-band signals as well as many imaging applications.

An optical fiber usually consists of a cylindrical ***fiber core*** with an index of refraction n_f, surrounded by another cylinder of lower index of refraction, n_c, called the ***cladding*** (Fig. 8-12(b)). The cladding layer serves to optically isolate the fiber when a large number of fibers are packed in close proximity, thereby avoiding leakage of light from one fiber into another. To ensure total internal reflection, the incident angle θ_3 in the fiber core must be equal to or greater than the critical angle θ_c for a wave in the fiber medium (with n_f) incident upon the cladding medium (with n_c). From Eq. (8.32a), we have

$$\sin\theta_c = \frac{n_c}{n_f}. \tag{8.35}$$

To meet the total reflection requirement $\theta_3 \geq \theta_c$, it is necessary that $\sin\theta_3 \geq n_c/n_f$. The angle θ_2 is the complement of angle θ_3; hence, $\cos\theta_2 = \sin\theta_3$. The necessary condition therefore may be written as

$$\cos\theta_2 \geq \frac{n_c}{n_f}. \tag{8.36}$$

Moreover, θ_2 is related to the incidence angle on the face of the fiber, θ_i, by Snell's law:

$$\sin\theta_2 = \frac{n_0}{n_f}\sin\theta_i, \tag{8.37}$$

where n_0 is the index of refraction of the medium surrounding the fiber ($n_0 = 1$ for air and $n_0 = 1.33$ if the fiber is in water) or

$$\cos\theta_2 = \left[1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i\right]^{1/2}. \tag{8.38}$$

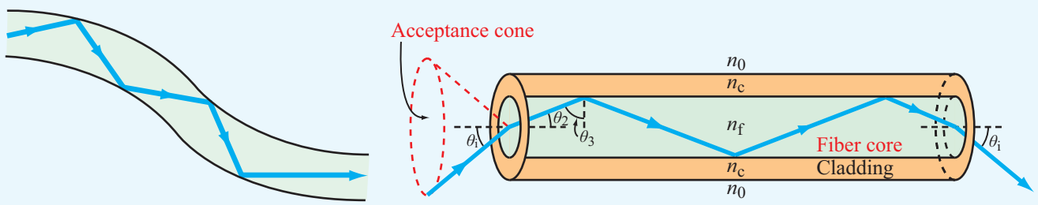
Using Eq. (8.38) on the left-hand side of Eq. (8.36) and then solving for $\sin\theta_i$ gives

$$\sin\theta_i \leq \frac{1}{n_0}(n_f^2 - n_c^2)^{1/2}. \tag{8.39}$$

8-3.1 Maximum Acceptance Cone

The ***acceptance angle*** θ_a is defined as the maximum value of θ_i for which the condition of total internal reflection remains satisfied:

$$\sin\theta_a = \frac{1}{n_0}(n_f^2 - n_c^2)^{1/2}. \tag{8.40}$$



(a) Optical fiber (b) Successive internal reflections

Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

因此，$\theta_3 = \theta_1$。板移动了光束的位置，但光束的方向保持不变。

练习8-4：在电磁波谱的可见光部分，水的折射率为1.33。向上看的水下光源产生的光波的临界角是多少？

答案：$\theta_c = 48.8°$。（参见EM。）

练习8-5：如果练习8-4的光源位于水面以下1m的深度，并且其光束是各向同性的（向各个方向辐射），那么从上方观察时，它会照亮多大的圆？

答案：圆的直径= 2.28 m。（参见EM。）

8-3 光纤

通过连续的全内反射，如图 8-12(a) 所示，光可以被引导通过由玻璃或透明塑料（称为光纤）制成的细介电棒。由于光被限制在棒内传播，唯一的功率损失是由于光纤发送端和接收端的反射以及光纤材料的吸收（因为它不是理想的电介质）。光纤可用于传输宽带信号以及许多成像应用。

光纤通常由折射率为 n_f 的圆柱形纤芯组成，周围环绕着另一个折射率较低的圆柱体 n_c，称为包层（图 8-12(b)）。当大量光纤紧密排列时，包层用于光学隔离光纤，从而避免光从一根光纤泄漏到另一根光纤中。为了确保全内反射，纤芯中的入射角 θ_3 必须等于或大于光纤介质中的波（n_f）入射到包层介质（n_c）上的临界角 θ_c。从方程。（8.32a），我们有

$$\sin\theta_c = \frac{n_c}{n_f}. \tag{8.35}$$

为了满足总反射要求$\theta_3 \geq \theta_c$，必须有$\sin\theta_3 \geq n_c/n_f$。角$\theta_2$是角$\theta_3$的余角；因此，$\cos\theta_2 = \sin\theta_3$。因此必要条件可以写为

$$\cos\theta_2 \geq \frac{n_c}{n_f}. \tag{8.36}$$

此外，根据斯涅耳定律，θ_2 与光纤表面的入射角 θ_i 相关：

$$\sin\theta_2 = \frac{n_0}{n_f}\sin\theta_i, \tag{8.37}$$

其中 n_0 是光纤周围介质的折射率（对于空气，$n_0 = 1$，如果光纤在水中，则 $n_0 = 1.33$）或

$$\cos\theta_2 = \left[1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i\right]^{1/2}. \tag{8.38}$$

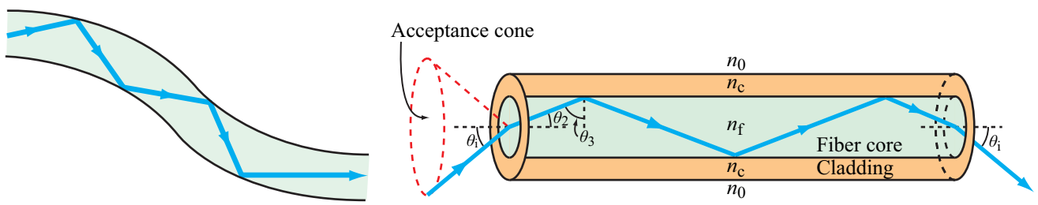
使用方程式（8.38）在等式的左侧。(8.36) 然后求解 $\sin\theta_i$ 给出

$$\sin\theta_i \leq \frac{1}{n_0}(n_f^2 - n_c^2)^{1/2}. \tag{8.39}$$

8-3.1 最大接收锥角

接受角 θ_a 定义为满足全内反射条件的 θ_i 的最大值：

$$\sin\theta_a = \frac{1}{n_0}(n_f^2 - n_c^2)^{1/2}. \tag{8.40}$$



(a) Optical fiber (b) Successive internal reflections

菲图8-12 只要反射角超过全内反射的临界角，波就可以沿着光纤传导。

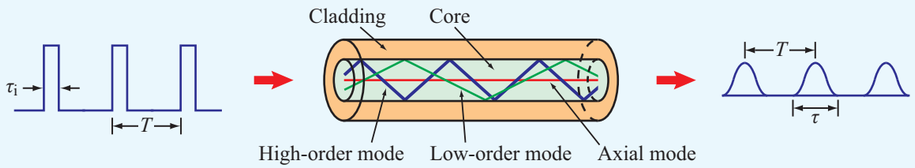


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

The angle θ_a is equal to half the angle of the acceptance cone of the fiber. Any ray of light incident upon the face of the core fiber at an incidence angle within the acceptance cone can propagate down the core. This means that there can be a large number of ray paths, called ***modes***, by which light energy can travel in the core. Rays characterized by large angles θ_i travel longer paths than rays that propagate along the axis of the fiber, as illustrated by the three modes shown in **Fig. 8-13**. Consequently, different modes have different transit times between the two ends of the fiber. This property of optical fibers is called ***modal dispersion*** and has the undesirable effect of changing the shape of pulses used for the transmission of digital data. When a rectangular pulse of light incident upon the face of the fiber gets broken up into many modes and the different modes do not arrive at the other end of the fiber at the same time, the pulse gets distorted—both in shape and length. In the example shown in **Fig. 8-13**, the narrow rectangular pulses at the input side of the optical fiber are of width τ_i separated by a time duration T. After propagating through the fiber core, modal dispersion causes the pulses to look more like spread-out sine waves with spread-out temporal width τ. If the output pulses spread out so much that $\tau > T$, the output signals will smear out, making it impossible to decipher the transmitted message from the output signal. Hence, to ensure that the transmitted pulses remain distinguishable at the output side of the fiber, it is necessary that τ be shorter than T. As a safety margin, it is common practice to design the transmission system such that $T \geq 2\tau$.

The spread-out width τ is equal to the time delay Δt between the arrival of the slowest ray and the arrival of the fastest ray. The slowest ray is the one traveling the longest distance and corresponds to the ray incident upon the input face of the fiber at the acceptance angle θ_a. From the geometry in **Fig. 8-12(b)** and Eq. (8.36), this ray corresponds to $\cos\theta_2 = n_c/n_f$. For an optical fiber of length l, the length of the path traveled by such a ray is

$$l_{max} = \frac{l}{\cos\theta_2} = l\,\frac{n_f}{n_c}, \tag{8.41}$$

and its travel time in the fiber at velocity $u_p = c/n_f$ is

$$t_{max} = \frac{l_{max}}{u_p} = \frac{ln_f^2}{cn_c}. \tag{8.42}$$

The minimum time of travel is realized by the axial ray and is given by

$$t_{min} = \frac{l}{u_p} = \frac{l}{c}\,n_f. \tag{8.43}$$

The total time delay is therefore

$$\tau = \Delta t = t_{max} - t_{min} = \frac{ln_f}{c}\left(\frac{n_f - 1}{n_c}\right) \quad \text{(s)}. \tag{8.44}$$

As we stated before, to retrieve the desired information from the transmitted signals, it is advisable that T, which is the interpulse period of the input train of pulses, be no shorter than 2τ. This in turn means that the ***maximum data rate*** (in bits per second), or equivalently the number of pulses per second, that can be transmitted through the fiber is limited to

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{cn_c}{2ln_f(n_f - n_c)} \quad \text{(bits/s)}. \tag{8.45}$$

Example 8-5: Transmission Data Rate on Optical Fibers

A 1-km long optical fiber (in air) is made of a fiber core with an index of refraction of 1.52 and a cladding with an index of refraction of 1.49. Determine

(a) the acceptance angle θ_a and

(b) the maximum usable data rate of signals that can be transmitted through the fiber.

Solution: **(a)** From Eq. (8.40),

$$\sin\theta_a = \frac{1}{n_0}(n_f^2 - n_c^2)^{1/2} = [(1.52)^2 - (1.49)^2]^{1/2} = 0.3,$$

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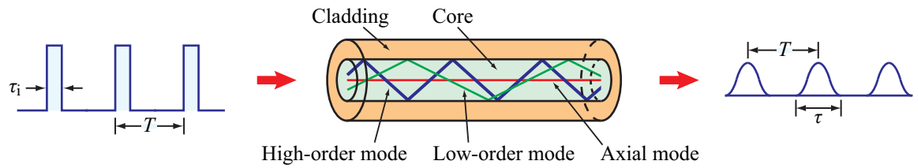


图 8-13 光纤中模色散引起的矩形脉冲失真。

角度θ_a等于光纤接收锥角的一半。任何以接收锥内的入射角入射到纤芯光纤表面的光线都可以沿着纤芯传播。这意味着可以有大量的光线路径（称为模式），光能可以通过这些路径在核心中传播。以大角度 θ_i 为特征的光线比沿光纤轴传播的光线传播的路径更长，如图 8-13 中的三种模式所示。因此，不同的模式在光纤两端之间具有不同的传输时间。光纤的这种特性称为模式色散，会产生改变用于传输数字数据的脉冲形状的不良影响。当入射到光纤表面的矩形光脉冲被分解成多种模式并且不同模式不同时到达光纤的另一端时，脉冲在形状和长度上都会发生扭曲。在图 8-13 所示的示例中，光纤输入侧的窄矩形脉冲的宽度为 τ_i，间隔时间为 T。通过光纤纤芯传播后，模态色散使脉冲看起来更像具有扩展时间宽度 τ 的扩展正弦波。如果输出脉冲扩散得如此之大，以至于 $\tau > T$，输出信号将变得模糊，从而无法从输出信号中破译传输的消息。因此，为了确保传输的脉冲在光纤的输出侧保持可区分，τ 必须短于 T。作为安全裕度，通常的做法是将传输系统设计为 $T \geq 2\tau$。

展开宽度τ等于最慢射线到达和最快射线到达之间的时间延迟Δt。最慢的光线是传播最长距离的光线，对应于以接收角 θ_a 入射到光纤输入面上的光线。从图 8-12(b) 和方程中的几何形状可以看出：(8.36)，该射线对应于 $\cos\theta_2 = n_c/n_f$。对于长度为 l 的光纤，该光线传播的路径长度为

$$l_{max} = \frac{l}{\cos\theta_2} = l\,\frac{n_f}{n_c}, \tag{8.41}$$

其在光纤中以速度 $u_p = c/n_f$ 的传播时间为

$$t_{max} = \frac{l_{max}}{u_p} = \frac{ln_f^2}{cn_c}. \tag{8.42}$$

最小传播时间由轴向射线实现，由下式给出

$$t_{min} = \frac{l}{u_p} = \frac{l}{c}\,n_f. \tag{8.43}$$

因此总延迟时间为

$$\tau = \Delta t = t_{max} - t_{min} = \frac{ln_f}{c}\left(\frac{n_f - 1}{n_c}\right) \quad \text{(s)}. \tag{8.44}$$

正如我们之前所说，为了从传输的信号中检索所需的信息，建议输入脉冲串的脉冲间周期 T 不小于 2τ。这又意味着可以通过光纤传输的最大数据速率（以每秒位数为单位）或等效的每秒脉冲数限制为

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{cn_c}{2ln_f(n_f - n_c)} \quad \text{(位/秒)}。 \tag{8.45}$$

例 8-5： 光纤上的传输数据速率

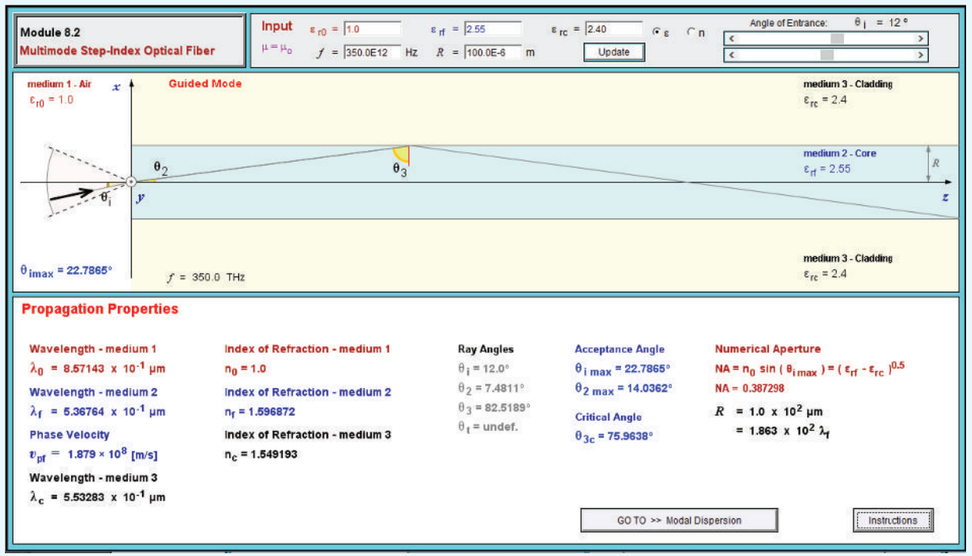
1 公里长的光纤（在空气中）由折射率为 1.52 的纤芯和折射率为 1.49 的包层组成。决定

(a) 接收角 θ_a 和 (b) 可通过光纤传输的信号的最大可用数据速率。

解： (a) 根据方程。 (8.40),

$$\sin\theta_a = \frac{1}{n_0}(n_f^2 - n_c^2)^{1/2} = [(1.52)^2 - (1.49)^2]^{1/2} = 0.3,$$

Module 8.2 Multimode Step-Index Optical Fiber Choose the indices on the fiber core and cladding and then observe the zigzag pattern of the wave propagation inside the fiber.



which corresponds to $\theta_a = 17.5°$.

(b) From Eq. (8.45),

$$f_p = \frac{cn_c}{2ln_f(n_f - n_c)} = \frac{3 \times 10^8 \times 1.49}{2 \times 10^3 \times 1.52(1.52 - 1.49)} = 4.9 \text{ (Mb/s)}.$$

8-3.2 Confined Acceptance Cone

The expression for the data rate given by Eq. (8.45) pertains to the case where the light coupled to the fiber is confined to the acceptance cone defined by the acceptance angle θ_a given by Eq. (8.40). If the entrance cone is larger than the acceptance cone, then some of the light will leak out of the fiber through the cladding, which not only reduces the power carried by the light but may also cause interference with signals carried by neighboring fibers.

On the other hand, if the entrance cone can be made to be smaller than the acceptance cone, the data rate can be safely increased. Let us denote $\theta_i(\max)$ as the maximum angle of the entrance cone. Per Eq. (8.38), $\theta_2(\max)$ is related to $\theta_i(\max)$ by

$$\cos\theta_2(\max) = \left[1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i(\max)\right]^{1/2}.$$

Repetition of the steps between Eqs. (8.41) and (8.45) leads to

$$f_p = \frac{c}{2ln_f}\left[\frac{\cos\theta_2(\max)}{1 - \cos\theta_2(\max)}\right] \qquad (\theta_i(\max) < \theta_a).$$

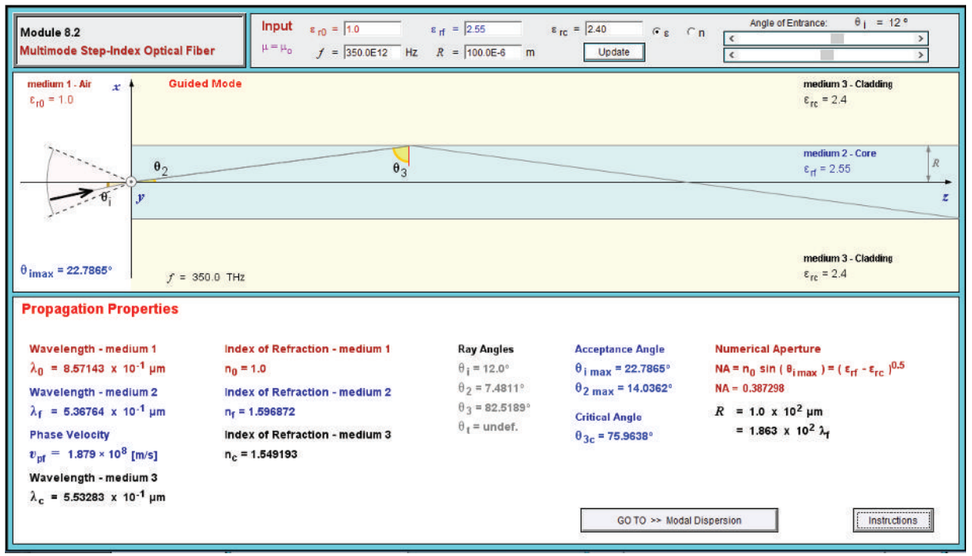
Exercise 8-6: If the index of refraction of the cladding material in Example 8-5 is increased to 1.50, what would be the new maximum usable data rate?

Answer: 7.4 (Mb/s). (See EM.)

8-4 Wave Reflection and Transmission at Oblique Incidence

In this section, we develop a rigorous theory of reflection and refraction of plane waves obliquely incident upon planar

模块 8.2 多模阶跃折射率光纤 选择纤芯和包层的折射率，然后观察光纤内波传播的锯齿形图案。



对应于 $\theta_a = 17.5°$。

(b) 从方程式(8.45),

$$f_p = \frac{cn_c}{2ln_f(n_f - n_c)} = \frac{3 \times 10^8 \times 1.49}{2 \times 10^3 \times 1.52(1.52 - 1.49)} = 4.9 \text{ (Mb/s)}.$$

8-3.2 受限接受锥

数据速率的表达式由方程式给出。（8.45）适用于耦合到光纤的光被限制在由方程（8.45）给出的接收角θ_a定义的接收锥内的情况。（8.40）。如果入口锥体大于接收锥体，则一些光将通过包层泄漏出光纤，这不仅会降低光携带的功率，而且还可能对相邻光纤携带的信号造成干扰。

另一方面，如果可以使入口锥体小于接收锥体，则可以安全地提高数据速率。让我们将 $\theta_i(\max)$ 表示为最大值入口锥体的角度。根据方程式(8.38)，$\theta_2(\max)$ 与 $\theta_i(\max)$ 的关系为

$$\cos\theta_2(\max) = \left[1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i(\max)\right]^{1/2}.$$

重复方程式之间的步骤。(8.41) 和 (8.45) 领先 到

$$f_p = \frac{c}{2ln_f}\left[\frac{\cos\theta_2(\max)}{1 - \cos\theta_2(\max)}\right] \qquad (\theta_i(\max) < \theta_a).$$

练习8-6：如果示例8-5中的包层材料的折射率增加到1.50，新的最大可用数据速率是多少？

答案：7.4（Mb/s）。（参见EM。）

8-4 斜入射波的反射和透射

在本节中，我们发展了一个关于倾斜入射到平面上的平面波的反射和折射的严格理论。

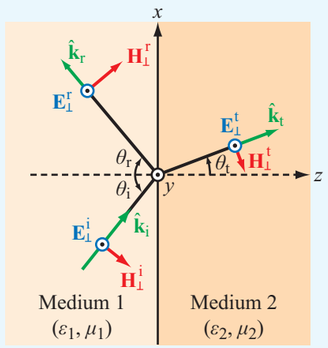
boundaries between different media. Our treatment parallels that in Section 8-1 for the normal-incidence case and goes beyond that in Section 8-2 on Snell's laws, which yielded information on only the angles of reflection and refraction.

For normal incidence, the reflection and transmission coefficients Γ and τ at a boundary between two media are independent of the polarization of the incident wave, because both the electric and magnetic fields of a normally incident plane wave are tangential to the boundary regardless of the wave polarization. This is not the case for obliquely incident waves traveling at an angle $\theta_i \neq 0$ with respect to the normal to the interface.

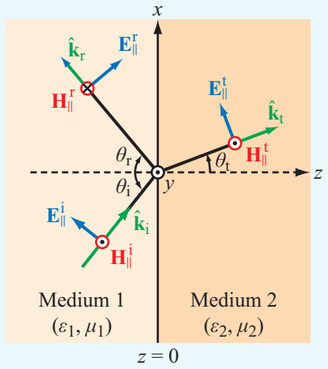
► The ***plane of incidence*** is defined as the plane containing the normal to the boundary and the direction of propagation of the incident wave. ◄

A wave of arbitrary polarization may be described as the superposition of two orthogonally polarized waves: one with its electric field parallel to the plane of incidence (***parallel polarization***) and the other with its electric field perpendicular to the plane of incidence (***perpendicular polarization***). These two polarization configurations are shown in **Fig. 8-14**, where the plane of incidence is coincident with the x–z plane. Polarization with **E** perpendicular to the plane of incidence is also called ***transverse electric*** (***TE***) polarization because **E** is perpendicular to the plane of incidence and with **E** parallel to the plane of incidence is called ***transverse magnetic*** (***TM***) polarization because in that case it is the magnetic field that is perpendicular to the plane of incidence.

For the general case of a wave with an arbitrary polarization, it is common practice to decompose the incident wave $(\mathbf{E}^i, \mathbf{H}^i)$ into a perpendicularly polarized component $(\mathbf{E}^i_\perp, \mathbf{H}^i_\perp)$ and a parallel polarized component $(\mathbf{E}^i_\parallel, \mathbf{H}^i_\parallel)$. Then, after determining the reflected waves $(\mathbf{E}^r_\perp, \mathbf{H}^r_\perp)$ and $(\mathbf{E}^r_\parallel, \mathbf{H}^r_\parallel)$ due to the two incident components, the reflected waves are added together to give the total reflected wave $(\mathbf{E}^r, \mathbf{H}^r)$ corresponding to the original incident wave. A similar process can be used to determine the total transmitted wave $(\mathbf{E}^t, \mathbf{H}^t)$.



(a) Perpendicular polarization



(b) Parallel polarization

Figure 8-14 The plane of incidence is the plane containing the direction of wave travel, $\hat{\mathbf{k}}_i$, and the surface normal to the boundary. In the present case, the plane of incidence containing $\hat{\mathbf{k}}_i$ and $\hat{\mathbf{z}}$ coincides with the plane of the paper. A wave is (a) perpendicularly polarized when its electric field vector is perpendicular to the plane of incidence and (b) parallel polarized when its electric field vector lies in the plane of incidence.

8-4.1 Perpendicular Polarization

Figure 8-15 shows a perpendicularly polarized incident plane wave propagating along the x_i direction in dielectric medium 1. The electric field phasor $\widetilde{\mathbf{E}}^i_\perp$ points along the y direction, and the associated magnetic field phasor $\widetilde{\mathbf{H}}^i_\perp$ is along the y_i axis. The directions of $\widetilde{\mathbf{E}}^i_\perp$ and $\widetilde{\mathbf{H}}^i_\perp$ are such that $\widetilde{\mathbf{E}}^i_\perp \times \widetilde{\mathbf{H}}^i_\perp$ points along the propagation direction $\hat{\mathbf{x}}_i$. The electric and magnetic fields of such a plane wave are given by

$$\widetilde{\mathbf{E}}^i_\perp = \hat{\mathbf{y}} E^i_{\perp 0} e^{-jk_1 x_i}, \tag{8.46a}$$

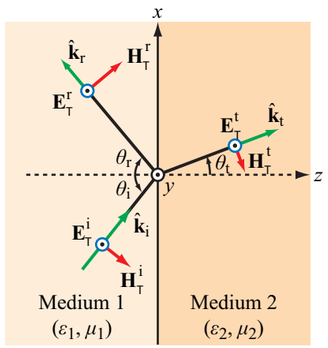
$$\widetilde{\mathbf{H}}^i_\perp = \hat{\mathbf{y}}_i \frac{E^i_{\perp 0}}{\eta_1} e^{-jk_1 x_i}, \tag{8.46b}$$

不同媒体之间的界限。我们的处理方法与第 8-1 节中针对正入射情况的处理方法相似，并且超出了第 8-2 节中关于斯涅耳定律的处理方法，后者仅提供了有关反射和折射角度的信息。

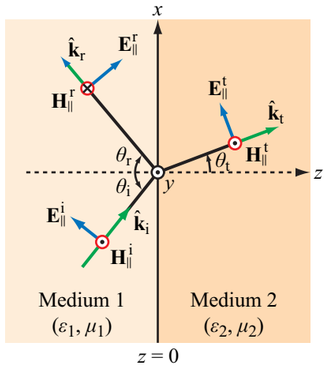
对于法向入射，两种介质之间边界处的反射和透射系数 Γ 和 τ 与入射波的偏振无关，因为无论波偏振如何，法向入射平面波的电场和磁场都与边界相切。对于相对于界面法线以角度$\theta_i \neq 0$ 传播的倾斜入射波，情况并非如此。

► 入射平面定义为包含边界法线和入射波传播方向的平面。 ◄

任意偏振波可以描述为两个正交偏振波的叠加：一个的电场平行于入射平面（平行偏振），另一个的电场垂直于入射平面（垂直偏振）。这两种偏振配置如图 8-14 所示，其中入射平面与 x-z 平面重合。 E 垂直于入射平面的偏振也称为横向电 (TE) 偏振，因为 E 垂直于入射平面，而 E 平行于入射平面的偏振称为横向磁 (TM) 偏振，因为在这种情况下，磁场垂直于入射平面。



(a) Perpendicular polarization



(b) Parallel polarization

图 8-14 入射平面是包含波传播方向 $\hat{k}_i$ 和垂直于边界的表面的平面。在本例中，包含k_i和z的入射平面与纸平面重合。当波的电场矢量垂直于入射平面时，波是（a）垂直偏振的； （b）当其电场矢量位于入射平面时，波是平行偏振的。

对于具有任意偏振的波的一般情况，通常的做法是将入射波 (E^i, H^i) 分解为垂直偏振分量 $(E^i_\perp, H^i_\perp)$ 和平行偏振分量 $(E^i_\parallel, H^i_\parallel)$。然后，在确定由两个入射分量引起的反射波（$E^r_\perp$，$H^r_\perp$）和（$E^r_\parallel$，$H^r_\parallel$）后，将反射波相加，得到与原始入射波相对应的总反射波（E^r，H^r）。可以使用类似的过程来确定总透射波（E^t，H^t）。

8-4.1 垂直偏振

图 8-15 显示了在电介质 1 中沿 x_i 方向传播的垂直偏振入射平面波。电场相量 $\tilde{}E^i_\perp$ 沿 y 方向指向，相关的磁场相量 $\tilde{}H^i_\perp$ 沿 y_i 轴指向。 $\tilde{}E^i_\perp$ 和 $\tilde{}H^i_\perp$ 的方向使得 $\tilde{}E^i_\perp \times \tilde{}H^i_\perp$ 沿着传播方向 $\hat{x}_i$ 指向。电和磁

这种平面波的场由下式给出

$$\widetilde{\mathbf{E}}^i_\perp = \hat{\mathbf{y}} E^i_{\perp 0} e^{-jk_1 x_i}, \tag{8.46a}$$

$$\widetilde{\mathbf{H}}^i_\perp = \hat{\mathbf{y}}_i \frac{E^i_{\perp 0}}{\eta_1} e^{-jk_1 x_i}, \tag{8.46b}$$

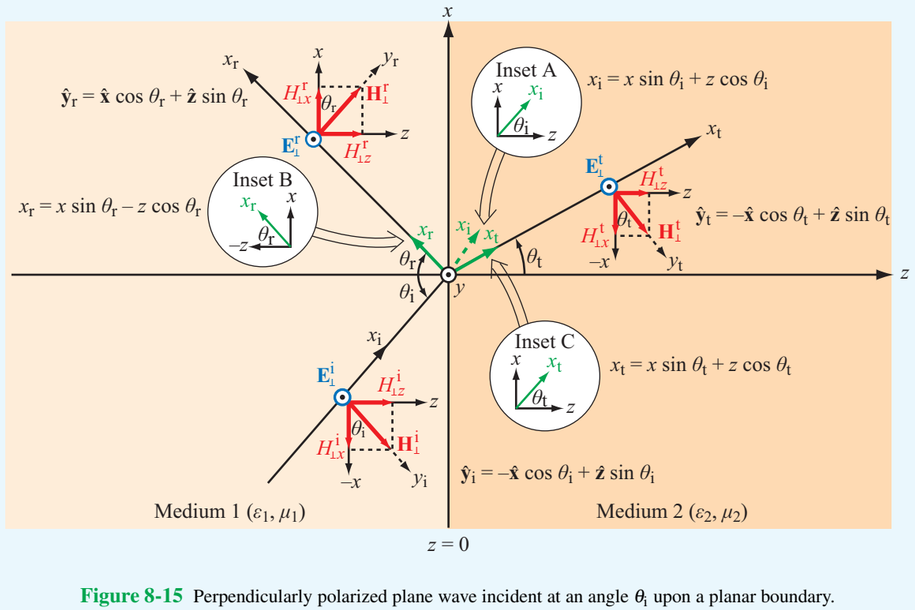


Figure 8-15 Perpendicularly polarized plane wave incident at an angle θ_i upon a planar boundary.

where $E^i_{\perp 0}$ is the amplitude of the electric field phasor at $x_i = 0$ and $k_1 = \omega\sqrt{\mu_1\varepsilon_1}$ and $\eta_1 = \sqrt{\mu_1/\varepsilon_1}$ are the wavenumber and intrinsic impedance of medium 1. From **Fig. 8-15**, the distance x_i and the unit vector $\hat{\mathbf{y}}_i$ may be expressed in terms of the (x, y, z) global coordinate system as

$$x_i = x\sin\theta_i + z\cos\theta_i, \tag{8.47a}$$

$$\hat{\mathbf{y}}_i = -\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{z}}\sin\theta_i. \tag{8.47b}$$

Substituting Eqs. (8.47a) and (8.47b) into Eqs. (8.46a) and (8.46b) gives

Incident Wave

$$\widetilde{\mathbf{E}}^i_\perp = \hat{\mathbf{y}} E^i_{\perp 0} e^{-jk_1(x\sin\theta_i + z\cos\theta_i)}, \tag{8.48a}$$

$$\widetilde{\mathbf{H}}^i_\perp = (-\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{z}}\sin\theta_i)\frac{E^i_{\perp 0}}{\eta_1} e^{-jk_1(x\sin\theta_i + z\cos\theta_i)}. \tag{8.48b}$$

With the aid of the directional relationships given in **Fig. 8-15** for the reflected and transmitted waves, these fields are given by the following:

Reflected Wave

$$\widetilde{\mathbf{E}}^r_\perp = \hat{\mathbf{y}} E^r_{\perp 0} e^{-jk_1 x_r} = \hat{\mathbf{y}} E^r_{\perp 0} e^{-jk_1(x\sin\theta_r - z\cos\theta_r)}, \tag{8.49a}$$

$$\widetilde{\mathbf{H}}^r_\perp = \hat{\mathbf{y}}_r \frac{E^r_{\perp 0}}{\eta_1} e^{-jk_1 x_r} = (\hat{\mathbf{x}}\cos\theta_r + \hat{\mathbf{z}}\sin\theta_r)\frac{E^r_{\perp 0}}{\eta_1} e^{-jk_1(x\sin\theta_t - z\cos\theta_t)}, \tag{8.49b}$$

Transmitted Wave

$$\widetilde{\mathbf{E}}^t_\perp = \hat{\mathbf{y}} E^t_{\perp 0} e^{-jk_2 x_t} = \hat{\mathbf{y}} E^t_{\perp 0} e^{-jk_2(x\sin\theta_t + z\cos\theta_t)}, \tag{8.49c}$$

$$\widetilde{\mathbf{H}}^t_\perp = \hat{\mathbf{y}}_t \frac{E^t_{\perp 0}}{\eta_2} e^{-jk_2 x_t} = (-\hat{\mathbf{x}}\cos\theta_t + \hat{\mathbf{z}}\sin\theta_t)\frac{E^t_{\perp 0}}{\eta_2} e^{-jk_2(x\sin\theta_t + z\cos\theta_t)}, \tag{8.49d}$$

where θ_r and θ_t are the reflection and transmission angles shown in **Fig. 8-15** and k_2 and η_2 are the wavenumber and intrinsic impedance of medium 2. Our goal is to describe the reflected and transmitted fields in terms of the parameters that characterize the incident wave, namely the incidence angle θ_i and the amplitude $E^i_{\perp 0}$. The four expressions given by Eqs. (8.49a) through (8.49d) contain four unknowns: $E^r_{\perp 0}$, $E^t_{\perp 0}$, θ_r, and θ_t. Even though angles θ_r and θ_t are related to θ_i by Snell's laws (Eqs. (8.28a) and (8.28b)), here we choose to

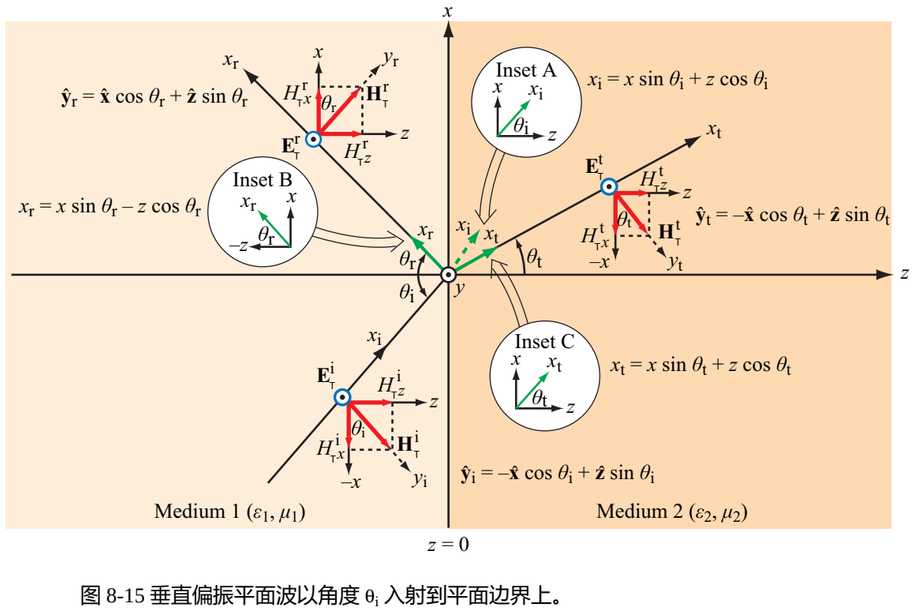


图 8-15 垂直偏振平面波以角度 θ_i 入射到平面边界上。

其中$E^i_{\perp 0}$是$x_i = 0$处电场相量的幅度，$k_1 = \omega\sqrt{\mu_1\varepsilon_1}$和$\eta_1 = \sqrt{\mu_1/\varepsilon_1}$是介质1的波数和本征阻抗。从图8-15中，距离$x_i$和单位向量$\hat{y}_i$可以用(x,y,z)全局坐标系表示为

$$x_i = x\sin\theta_i + z\cos\theta_i, \tag{8.47a}$$

$$\hat{\mathbf{y}}_i = -\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{z}}\sin\theta_i. \tag{8.47b}$$

代入等式。（8.47a）和（8.47b）代入等式。(8.46a) 和 (8.46b) 给出

事件波

$$\widetilde{\mathbf{E}}^i_\perp = \hat{\mathbf{y}} E^i_{\perp 0} e^{-jk_1(x\sin\theta_i + z\cos\theta_i)}, \tag{8.48a}$$

$$\widetilde{\mathbf{H}}^i_\perp = (-\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{z}}\sin\theta_i)\frac{E^i_{\perp 0}}{\eta_1} e^{-jk_1(x\sin\theta_i + z\cos\theta_i)}. \tag{8.48b}$$

借助图 8-15 中给出的反射波和透射波的方向关系，这些场由下式给出：

反射波

$$\widetilde{\mathbf{E}}^r_\perp = \hat{\mathbf{y}} E^r_{\perp 0} e^{-jk_1 x_r} = \hat{\mathbf{y}} E^r_{\perp 0} e^{-jk_1(x\sin\theta_r - z\cos\theta_r)}, \tag{8.49a}$$

$$\widetilde{\mathbf{H}}^r_\perp = \hat{\mathbf{y}}_r \frac{E^r_{\perp 0}}{\eta_1} e^{-jk_1 x_r} = (\hat{\mathbf{x}}\cos\theta_r + \hat{\mathbf{z}}\sin\theta_r)\frac{E^r_{\perp 0}}{\eta_1} e^{-jk_1(x\sin\theta_t - z\cos\theta_t)}, \tag{8.49b}$$

传输波

$$\widetilde{\mathbf{E}}^t_\perp = \hat{\mathbf{y}} E^t_{\perp 0} e^{-jk_2 x_t} = \hat{\mathbf{y}} E^t_{\perp 0} e^{-jk_2(x\sin\theta_t + z\cos\theta_t)}, \tag{8.49c}$$

$$\widetilde{\mathbf{H}}^t_\perp = \hat{\mathbf{y}}_t \frac{E^t_{\perp 0}}{\eta_2} e^{-jk_2 x_t} = (-\hat{\mathbf{x}}\cos\theta_t + \hat{\mathbf{z}}\sin\theta_t)\frac{E^t_{\perp 0}}{\eta_2} e^{-jk_2(x\sin\theta_t + z\cos\theta_t)}, \tag{8.49d}$$

其中θ_r和θ_t是图8-15中所示的反射和透射角，k_2和η_2是介质2的波数和固有阻抗。我们的目标是用表征入射波的参数（即入射角θ_i和振幅$E^i_{\perp 0}$）来描述反射和透射场。等式给出的四个表达式。(8.49a) 到 (8.49d) 包含四个未知数：$E^r_{\perp 0}$、$E^t_{\perp 0}$、θ_r 和 θ_t。尽管根据斯涅耳定律，角度 θ_r 和 θ_t 与 θ_i 相关（方程（8.28a）和（8.28b））, 但这里我们选择